01 PROTOCOL

ECONOMIC CHARTER

FEE SCHEDULE · SYSTEM COSTS · REVENUE MODEL

v1.0.3 · TX-OPACITY RELEASE

Devnet live · Mainnet pending audit

Effective 2026-06-11 · Updated 2026-07-20

VOLTA TEAM · SLASHY FX · BUILT ON SOLANA

The numbers.

Protocol 01 is a privacy layer for Solana. Merchants integrate via SDK and receive payments at their own retailer wallet. The protocol takes a small fixed fee on shield and unshield. There are no platform commissions, no rev-share locks, no exclusivity clauses.

0.30% SHIELD FEE	**0.50%** UNSHIELD FEE	**0.80%** TOTAL ROUND-TRIP
0% SUBSCRIPTION CUT	**0%** MERCHANT SIGNUP	**5.00%** HARD CAP ON PROTOCOL FEE

What this document covers

What we do NOT charge

Free always

- Service registration on-chain
- Subscription vault creation
- Vault pause / resume
- Stealth address generation
- SDK access & API calls
- Push notifications, webhooks

Pass-through only

- Solana base tx fee — 5,000 lamports per signature
- Account rent — reclaimable on close
- Priority fees — user-controlled, market-driven
- RPC calls — merchant uses their own endpoint

Extension premium tiers — separate, off-chain

The Chrome extension sells Standard and ZK tiers through anonymous `p01lic_` license keys, verifiable via the merchant SDK without any PII. This is product revenue, not a protocol fee: it never touches the on-chain fee schedule above, and the free tier keeps full self-custody functionality.

Core protocol fees.

Three on-chain operations carry a fee. Every other operation is free. Fees are deducted in the same token as the operation (SOL or SPL), routed atomically to the protocol fee wallet, and recorded as part of the transaction signature.

Fee schedule

OPERATION	BPS	% RATE	WHEN CHARGED	SOURCE
SHIELD deposit into the shielded pool	30	0.30 %	At deposit, before pool credit	SHIELD_FEE_BPS
UNSHIELD withdraw from the shielded pool	50	0.50 %	At withdrawal, deducted from amount out	UNSHIELD_FEE_BPS
TRANSFER private 2-in-2-out shielded transfer	30	0.30 %	At each transfer, on the input notes	SHIELD_FEE_BPS
SUBSCRIBE lock note into a vault	0	0.00 %	Free — vault uses an existing shielded note	—
CLAIM retailer pulls a vault period	0	0.00 %	Free — the unshield at claim already taxed at 0.50%	—
CANCEL close vault, re-shield refund	0	0.00 %	Free — reshield uses existing pool, no new shield	—

Worked numbers

1 SOL one-shot payment

Buyer shields 1.000 SOL	-0.003 SOL
Net into pool	0.997 SOL
Subscription claim (no fee)	—
Retailer unshields 0.997 SOL	-0.005 SOL
Net to retailer	0.992 SOL
Effective take	0.80 %

50 SOL recurring vault

Subscriber shields 50.000 SOL	-0.150 SOL
Locked in vault	49.850 SOL
12 monthly claims at 4.154 SOL	12 × 0.021
Total unshield fee over year	-0.249 SOL
Net to retailer over year	49.601 SOL
Effective take	0.80 %

Fee floor for tiny amounts

For payments below 0.01 SOL, the 0.5% unshield fee can round to zero (0.5% × 0.01 = 50,000 lamports = 0.00005 SOL). The Solana base transaction fee (5,000 lamports per signature) becomes the dominant cost. Below ~0.001 SOL, the network fee exceeds the protocol fee — meaningful only for micropayments.

Solana network cost.

Every Solana operation carries a base transaction fee plus a compute budget for on-chain execution. Protocol 01 sets explicit compute unit limits per operation so users know the upper bound. These costs go to Solana validators — not to the protocol.

Compute units per operation

OPERATION	COMPUTE UNITS	MIN. PRIORITY FEE*	SOURCE
Shield	200,000	0.000005 SOL	COMPUTE_UNITS.SHIELD
Unshield (with STARK verify)	1,400,000	0.000035 SOL	COMPUTE_UNITS.STARK_VERIFY
Private transfer	400,000	0.000010 SOL	COMPUTE_UNITS.TRANSFER
Stealth send / claim	200,000	0.000005 SOL	COMPUTE_UNITS.STEALTH_*
Stream create	200,000	0.000005 SOL	COMPUTE_UNITS.STREAM_CREATE
Subscription create	300,000	0.000008 SOL	COMPUTE_UNITS.SUBSCRIPTION_CREATE
Instant unshield prefund	200,000	0.000005 SOL	PREFUND_CU

*Minimum priority fee assumes 25,000 micro-lamports per CU (a common devnet/mainnet baseline). Real fees fluctuate with network congestion; the SDK exposes a `priorityFeeMultiplier` knob so callers can scale.

Account rent — reclaimable

Some operations create new on-chain accounts which must be rent-exempt. The user funds the rent up-front and gets it back when the account is closed. None of this rent goes to the protocol.

ACCOUNT	SIZE	RENT	RECLAIMED BY
Shielded note nullifier PDA	~64 B	~0.00098 SOL	Permanent (anti double-spend)
Subscription vault	~190 B	~0.00146 SOL	Cancel returns rent
Service registry entry	~417 B	~0.00302 SOL	Deregister returns rent
STARK proof buffer	~120 KB	~0.85 SOL	Auto-closed at unshield (or on crash-sweep)
PQ stealth announcement (ML-KEM-768, V2)	~1,220 B	~0.0094 SOL	Close instruction planned (roadmap P2)
Liquidity pool prefund record	~80 B	~0.00111 SOL	Settle returns rent

Total network cost — typical 1 SOL unshield

0.005 SOL protocol fee + ~0.000035 SOL priority + 0.000005 SOL base. The proof buffer rent (~0.85 SOL) is a transient placeholder — auto-closed on the same transaction. The user sees one net debit of ~0.005045 SOL.

Liquidity pool fee.

Optional instant-unshield path: the liquidity pool prefunds the user's withdrawal so they get funds in the same block, in exchange for a small fee paid to LP providers. Standard unshield (free of LP fee) requires waiting for full proof finalization.

Two unshield paths

Standard unshield

- STARK proof generated on device (~6–10 s)
- Submitted to chain, verified, executed
- Funds at retailer wallet on next block
- Fee: 0.50% protocol, no LP fee

FREE OF LP FEE

Instant unshield (prefund)

- LP advances funds at retailer wallet immediately
- User completes the chain side asynchronously
- LP collects from the pool when proof lands
- Fee: 0.50% protocol + LP fee (variable, capped 3.00%)

UP TO 3.00% LP FEE

Liquidity pool parameters

PARAMETER	DEFAULT	MAXIMUM	SETTER
Prefund fee (per pool)	~0.50%	3.00%	LP authority via update_params
Min prefund amount	0.01 SOL	—	Pool init
Max prefund amount	100 SOL	—	Pool init
Settlement window	1 epoch	5 epochs	Pool init

Where the LP fee goes

80% LP providers (yield)	15% protocol	5% treasury

The LP fee compensates capital providers who lock liquidity for instant settlement. Eight in ten basis points flow to LPs as yield. The split is enforced on-chain at settlement time and is not negotiable per-trade.

When does instant unshield make sense?

SCENARIO	RECOMMENDED PATH	REASON
Subscription claim, retailer ops	STANDARD	Block-level latency is fine, save the LP fee
Live commerce checkout (~10 s budget)	INSTANT	UX matters more than 1–3% LP fee
High-value treasury sweep	STANDARD	3% on a large amount = real money
Mobile user, poor connectivity	INSTANT	Fall back to LP path on RPC retry / proof timeout

Mugen exchange fee.

Mugen is the reference fiat <> crypto P2P exchange built on Protocol 01. It is a separate product line with its own fee schedule. Merchants using only Protocol 01 SDK do not pay Mugen fees. Documented here so integrators know the full surface.

Mugen fee structure

Two-tier model: a visible fee shown at checkout, plus a small spread baked into the exchange rate. The spread covers FX risk and the fiat-rail counterparty premium. Total effective take varies by payment method.

METHOD	VISIBLE FEE	HIDDEN SPREAD	TOTAL EFFECTIVE	SETTLEMENT
SEPA / IBAN bank transfer	0.50 %	0.80 %	1.30 %	1–2 days
Wise	0.50 %	0.80 %	1.30 %	1–2 hours
Revolut	0.50 %	0.80 %	1.30 %	Instant
Cash (escrowed)	0.50 %	1.20 %	1.70 %	Same day
Card (Visa / Mastercard)	1.90 %	1.00 %	2.90 %	Instant

Why card is the highest tier

Card processors charge 1.5–2.0% interchange to merchants, plus chargeback exposure of typically 0.3–0.5% for crypto-adjacent flows. The 1.90% visible fee is set to break even on processor cost, not as protocol revenue. The 1.00% spread covers chargeback reserve and FX volatility on the card-to-SOL leg.

Revenue split — Mugen on-chain fees

The total on-chain Mugen fee is ~1.50% averaged across methods. At escrow release, this fee is split atomically between three wallets:

20% Protocol 01 | 65% Mugen ops | 15% Treasury

RECIPIENT	SHARE	USE
PROTOCOL 01	20 %	SDK, mobile, extension, infra hosting, audits
MUGEN	65 %	Fiat rail ops, escrow management, dispute mediation, support
TREASURY	15 %	DAO grants, governance staking, community programs

Layer 1 — Natural Variance Fiat

Mugen splits user payments into canonical SOL denominations (0.1 / 1 / 10 / 100) internally. The bank counterparty sees a natural-looking amount (e.g., €847.32). The chain sees a uniform pool entry. Residual fiat (< 0.50%) is tracked in a separate sub-wallet and reconciled in the next batch — not lost, not extra fee.

Where every basis point flows.

Single source of truth: the on-chain fee splitter program. Each fee type is routed to exactly one wallet, then sub-split atomically by the splitter. The protocol fee wallet address is hardcoded across all P-01 programs.

Single fee wallet (current)

```
PROTOCOL_FEE_WALLET = BRop3akxwuQaAHeMUC33ZyRjzLh78ENquVMgHum9TjNN
```

Hardcoded in `programs/p01-fee-splitter/src/lib.rs` and mirrored across `@protocol-01/privacy-sdk`. All shield / unshield / transfer fees converge here, then redistribute.

Distribution from the fee wallet

The fee splitter applies a fixed-ratio routing to four downstream accounts. Splits are configurable but capped on-chain:

50% Engineering & infra	25% Audits & security	15% Treasury	10% Founder

RECIPIENT	SHARE	MANDATE
ENGINEERING & INFRASTRUCTURE	50 %	Devs, RPC nodes, prover compute, relay nodes, CI / CD, mobile / extension upkeep
AUDIT & SECURITY	25 %	Recurring audits (OtterSec / Neodyme / Trail of Bits), bug bounties, formal verification
TREASURY RESERVE	15 %	12-month operating runway, emergency response, slashing escrow for relayer dispute
FOUNDER ALLOCATION	10 %	Volta Team / Slashy Fx — the sole party building the protocol

Why these splits

50% to engineering

Solana-native crypto infra is compute-expensive. The stack spans ~15 Anchor programs and ~340 tests. STARK proof generation runs on user devices but the verifier and prover state live on chain. The 50% cut keeps the build velocity sustainable through mainnet and beyond.

25% to audits

Privacy primitives without audit are theatre. We pre-allocate 25% of every basis point to recurring audits, not just a single pre-launch review. Mainnet deployment is gated on at least one complete audit cycle.

Worked example.

A SaaS subscription, monthly billing at €19.99 USDC equivalent, 12-month cohort of 1,000 paying users. End-to-end protocol economics, before the merchant's own COGS / refunds / churn assumptions.

Cohort assumptions

1,000	€19.99	12	10 %	€0.025	100 %
SUBS MONTH 1	MONTHLY PRICE	MONTHS TRACKED	MONTHLY CHURN	AVG NETWORK FEE	STANDARD PATH

Cash flow over 12 months

MONTH	ACTIVE SUBS	GROSS REVENUE	SHIELD FEE*	UNSHIELD FEE	NET TO MERCHANT
1	1,000	€19,990	€59.97	€99.95	€19,830.08
2	900	€17,991	€0	€89.96	€17,901.04
3	810	€16,192	€0	€80.96	€16,111.04
4	729	€14,573	€0	€72.86	€14,499.84
5	656	€13,116	€0	€65.58	€13,049.86
6	590	€11,805	€0	€59.02	€11,745.87
7	531	€10,624	€0	€53.12	€10,570.78
8	478	€9,562	€0	€47.81	€9,513.70
9	430	€8,606	€0	€43.03	€8,562.33
10	387	€7,745	€0	€38.73	€7,706.10
11	348	€6,971	€0	€34.85	€6,935.49
12	313	€6,274	€0	€31.37	€6,241.94
Total	–	€143,449	€59.97	€717.24	€142,668

*Shield fee is paid by the subscriber when funding their vault — once per subscription. Subsequent monthly claims hit only the unshield fee. Months 2-12 show €0 shield because no new vault is created.

Bottom line

€143,449	€777	€142,668
GROSS REVENUE	TOTAL FEES (0.54%)	NET TO MERCHANT

Compare: Stripe at 2.90% + €0.30 = ~€5,460 fee on the same revenue, no privacy.

Benchmark & governance.

Where Protocol 01 sits in the merchant-rail landscape, and the rules governing how these fees can change.

Comparison table

PROVIDER	TAKE RATE	PER-TX FIX	PRIVACY	SETTLEMENT
PROTOCOL 01	0.80 %	€0.025	NATIVE ZK	~10 s on-chain
Stripe (cards)	2.90 %	€0.30	NONE	2–7 days
Patreon	8–12 %	+ payout fees	NONE	Monthly
BTCPay self-host	0 %	on-chain BTC fee	L1 PSEUDONYMOUS	10–60 min
Coinbase Commerce	1.00 %	network fee	NONE	~30 min finality
Solana Pay (raw)	0 %	~€0.0001	NONE	~1 s on-chain

Protocol 01 sits between raw chain rails (zero fee, zero privacy) and fiat processors (high fee, no privacy). The premium of 0.80% over Solana Pay buys ZK privacy + STARK post-quantum + native subscription rails.

Fee change governance

MECHANISM	AUTHORITY	CONSTRAINT
Protocol fee BPS	Multisig (Volta core)	Hard cap 5.00% on-chain (MAX_FEE_BPS)
Fee wallet address	Multisig + 7-day timelock	Public announcement before activation
LP prefund fee per pool	Pool authority (per-LP)	Hard cap 3.00% on-chain (MAX_PREFUND_FEE_BPS)
Mugen fee split	Mugen team multisig	Splits sum-checked on-chain at every release
Splitter recipient ratios	Multisig + 30-day timelock	Treasury share cannot drop below 10% per protocol bylaws

Public commitments

We commit to

- Never raise the protocol fee above 1.00% without 30-day notice + community vote
- Publish quarterly fee revenue + recipient breakdown on-chain
- Open-source all fee accounting code & on-chain programs
- Honor pre-existing service agreements at the fee rate at signup

We do not

- Take any percentage of the merchant's actual revenue
- Charge platform / listing / verification fees
- Sell merchant data — we cannot, the chain hides it
- Enforce exclusivity — merchants can use other rails freely

One page summary.

Headline numbers

0.30 %	0.50 %	0.80 %
SHIELD	UNSHIELD	ROUND-TRIP

What you, the merchant, pay

- 0.30% on every shield (paid by subscriber, not by you)
- 0.50% on every unshield (deducted from amount you receive)
- ~0.000010 SOL Solana priority + 0.000005 SOL base per claim
- 0% merchant signup, 0% subscription cut, 0% platform commission
- Optional 0–3% LP fee if you opt into instant unshield path

What the system burns

- Compute units paid to Solana validators — 200K to 1.4M per op
- Account rent (mostly reclaimable on close) — 0.001–0.85 SOL per account
- RPC bandwidth — user / merchant pay their own RPC provider

What we, Protocol 01, take

- 0.30% shield + 0.50% unshield → protocol fee wallet
- 15% of LP fee on instant unshield → protocol fee wallet
- 20% of Mugen on-chain fee (only if Mugen rail used)
- Of which: 50% engineering, 25% audit, 15% treasury, 10% founder

Changelog — v1.0.3 (2026-07-20)

- Arcium MPC removed from Protocol 01 (module retired; Mugen governance only) — MPC vote row dropped from the compute table
- PQ stealth announcement account added to the rent table (ML-KEM-768 V2 transport, live on devnet; close instruction on the roadmap)
- Extension Standard / ZK license tiers documented as off-chain product revenue, zero PII

VOLTA TEAM · SLASHY FX

BUILT ON SOLANA · DEVNET LIVE · MAINNET PENDING AUDIT

PROTOCOL-01.DEV · V1.0.3 TX-OPACITY · 2026-07-20